

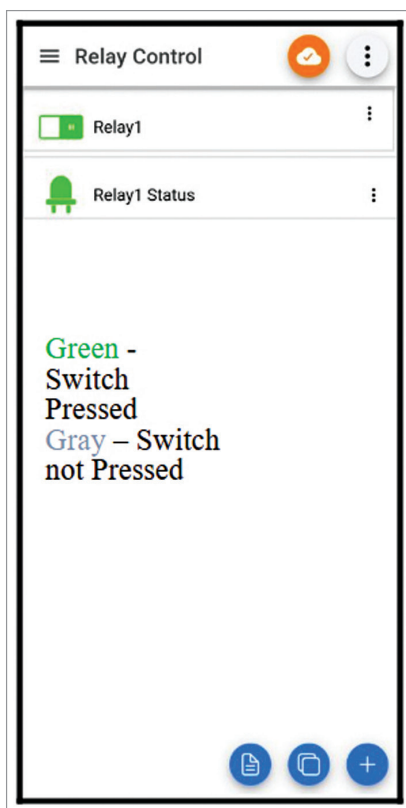


Fig. 12: Relay status

the same time, the relay status is published back to the MQTT broker, enabling the IoT application to display the current relay state. See Fig. 7.

After completing all the required configurations, the code can be compiled and uploaded to the ESP-01 module using the ESP-01 USB Programmer Adaptor. For other ESP8266 development boards, such as NodeMCU or WeMos D1 Mini, a separate programmer is not required because these boards already include an onboard USB-to-serial converter for direct programming through the USB port.

IoT MQTT panel app configuration

The IoT MQTT Panel app is a highly customisable IoT dashboard application based on the MQTT protocol. It enables users to visualise and control various inputs and outputs within an IoT application. The app is available on Android and iOS. The configuration steps are shown in Figs. 8 to 12.

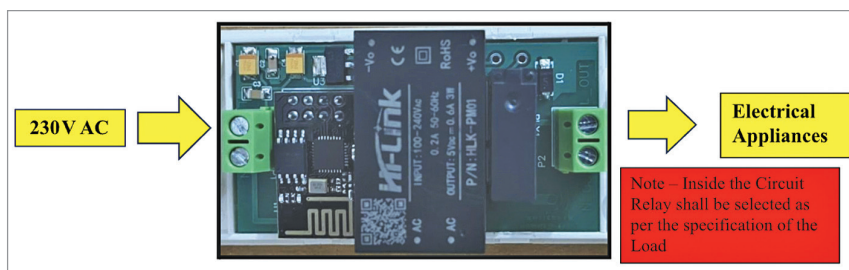


Fig. 13: Input and output connections for the appliance

Construction and testing

Assemble the circuit according to the circuit diagram shown in Fig. 2. The circuit includes an onboard HLK-PM01 module that converts a 230V AC mains supply to 5V DC. Therefore, appropriate precautions against electric shock must be taken while handling and testing the system.

After uploading the code to the ESP-01 and assembling the circuit, the prototype shown in Fig. 1 is ready for testing. Open the IoT MQTT panel app to control the relay through the MQTT broker. Once the ESP-01 successfully connects to the Wi-Fi network and MQTT server, the relay can be controlled directly from the application dashboard. The switch button in the app interface can be used to toggle the relay on and off remotely through MQTT communication.

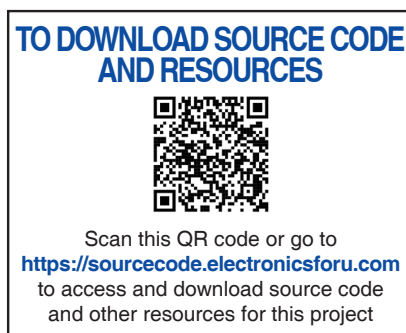
When the switch in the application is turned on, the MQTT app publishes the command 1 to the subscribed MQTT topic. The ESP-01 receives this command through the Switchsys MQTT broker and energises the relay via the ULN2003A relay driver circuit. Similarly, when the switch is turned off, the app

sends the command 0, and the ESP-01 de-energises the relay to turn the connected appliance off. The updated relay status is also published back to the MQTT broker so that the application dashboard can display the current relay state in real time.

After verifying proper operation through the app, the AC mains input and the electrical appliance can be connected to the relay terminals. The AC supply should be connected to the input terminal, while the electrical load or appliance should be connected to the control terminal, as shown in Fig. 13. Once connected, the appliance can be switched on and off wirelessly using the IoT MQTT panel application through the internet or a local Wi-Fi network, depending on MQTT connectivity.

Future scope

The system can be further enhanced with security features, feedback sensing, and automation scheduling, making it a practical and efficient IoT-based smart control solution. **EFY**



Pradipta Ghosh holds an M.E. in Electronics and Instrumentation. He is passionate about electronics, MCU-based embedded systems, and IoT devices.



Challa Naga Sai Ravali holds an M.Tech degree in Communication and Signal Processing. She is an embedded hardware design engineer with experience in embedded systems, FPGA-based design, IoT solutions, and telecom product development. Her areas of interest include hardware design, wireless communication, and connected technologies.